EXP NO: 02

EXPERIMENT 2: JPYTHON WIDGETS DASHBOARD

AIM

To create an interactive health and fitness dashboard using Python, Pandas, Matplotlib, and Jupyter Widgets for analyzing and visualizing wearable health telemetry data.

PROCEDURE

1. Import the required Python libraries.
2. Generate a synthetic health telemetry dataset using Python and NumPy.
3. Create interactive widgets for activity, step goal, chart type, and sleep threshold.
4. Filter the dataset based on the selected widget values.
5. Calculate summary health metrics.
6. Visualize the data using interactive Matplotlib charts.
7. Display the dashboard and update it dynamically based on user selections.

PROGRAM CODE

```
import numpy as np, pandas as pd, matplotlib.pyplot as plt, ipywidgets as widgets
from IPython.display import display, clear_output

np.random.seed(42)
n_days = 60
dates = pd.date_range(end=pd.Timestamp.now(), periods=n_days, freq="D")
activities = ["Running", "Cycling", "Swimming", "HIIT", "Walking"]

df = pd.DataFrame({
    "Date": dates,
    "Activity": np.random.choice(activities, n_days),
    "StepCount": np.random.randint(3000, 16000, n_days),
    "AvgHeartRate": np.random.randint(60, 165, n_days),
    "CaloriesBurned": np.random.randint(200, 900, n_days),
    "SleepScore": np.random.randint(50, 100, n_days)
})

activity = widgets.SelectMultiple(options=activities, value=activities,
```

```
description="Activities: ")
steps = widgets.IntSlider(value=5000, min=3000, max=12000, step=1000, description="Min
Steps Goal: ")
chart = widgets.RadioButtons(
    options=["Heart Rate vs Calories (Scatter)", "Daily Step Trends (Line)", "Sleep
Quality (Bar)"],
    value="Heart Rate vs Calories (Scatter)",
    description="Metric View: "
)
sleep_check = widgets.Checkbox(value=True, description="Highlight Optimal Sleep Range
(>=75)")

output = widgets.Output()

def update(change=None):
    with output:
        clear_output(wait=True)
        d = df[df.Activity.isin(activity.value) & (df.StepCount >= steps.value)]

        calories = d.CaloriesBurned.sum()
        hr = d.AvgHeartRate.mean() if len(d) else 0
        sleep = d.SleepScore.mean() if len(d) else 0

        print(f"--- HEALTH TELEMETRY SUMMARY ({len(d)} Days Matched Criteria) ---")
        print(f"Total Calories Burned: {calories} kcal | Avg Heart Rate: {hr:.1f} BPM |
Avg Sleep Score: {sleep:.1f}/100")
        print("=" * 85)

        fig, ax = plt.subplots(figsize=(10, 4.5))

        if chart.value == "Heart Rate vs Calories (Scatter)":
            if len(d):
                s = ax.scatter(d.AvgHeartRate, d.CaloriesBurned, c=d.SleepScore,
cmap="plasma", s=d.StepCount / 100, alpha=0.7, edgecolors="k")
                ax.set(title="Avg Heart Rate vs Calories Burned\n(Size = Steps, Color =
Sleep Score)", xlabel="Average Heart Rate (BPM)", ylabel="Calories Burned (kcal)")
                fig.colorbar(s, ax=ax).set_label("Sleep Score")
        elif chart.value == "Daily Step Trends (Line)":
            ax.plot(d.Date, d.StepCount, marker="o", linewidth=2)
            ax.axhline(steps.value, linestyle="--", label=f"Goal ({steps.value}
steps)")
            ax.set(title="Daily Step Count Progression", xlabel="Date", ylabel="Step
Count")
            ax.legend()
            plt.xticks(rotation=30)
        else:
            colors = ["#2ecc71" if x >= 75 else "#e74c3c" for x in d.SleepScore] if
sleep_check.value else "#3498db"
            ax.bar(d.Date.dt.strftime("%b %d"), d.SleepScore, color=colors)
            ax.set(title="Daily Sleep Quality Score", xlabel="Date", ylabel="Sleep
```

```
Score (0-100)")
            plt.xticks(rotation=45)

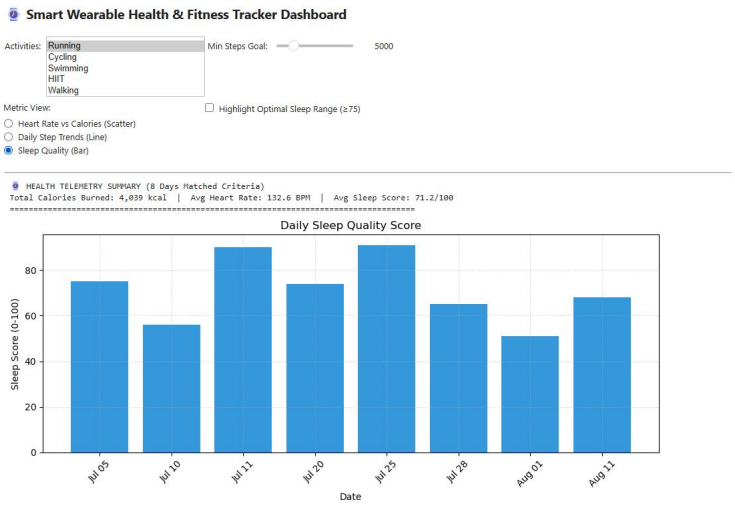
        ax.grid(True, linestyle=":", alpha=0.6)
        plt.tight_layout()
        plt.show()

for w in (activity, steps, chart, sleep_check):
    w.observe(update, names="value")

display(
    widgets.HTML("<h2>Smart Wearable Health & Fitness Tracker Dashboard</h2>"),
    widgets.HBox([activity, steps]),
    widgets.HBox([chart, sleep_check]),
    widgets.HTML("<hr>"),
    output
)

update()
```

OUTPUT



Smart Wearable Health & Fitness Tracker Dashboard

Activities: Running, Cycling, Swimming, HIIT, Walking — Min Steps Goal: 5000

Metric View:
- Heart Rate vs Calories (Scatter)
- Daily Step Trends (Line)
- Sleep Quality (Bar)

Highlight Optimal Sleep Range (≥75)

```
HEALTH TELEMETRY SUMMARY (8 Days Matched Criteria)
Total Calories Burned: 4,039 kcal  |  Avg Heart Rate: 132.6 BPM  |  Avg Sleep Score: 71.2/100
================================================================================
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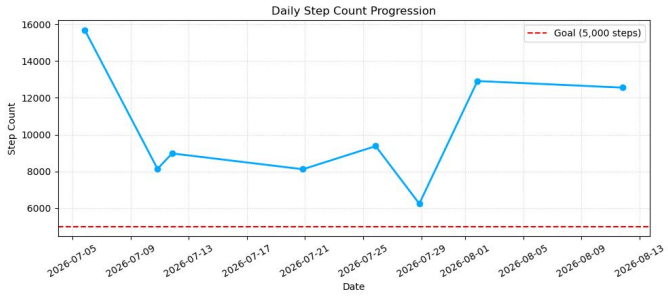
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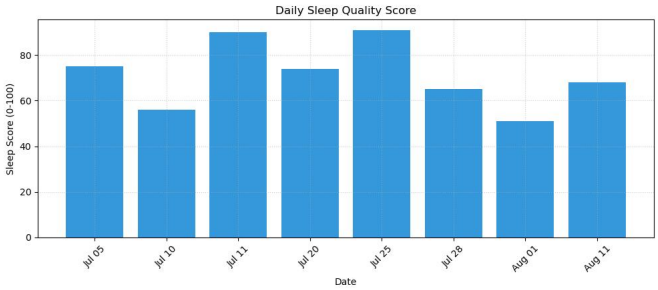
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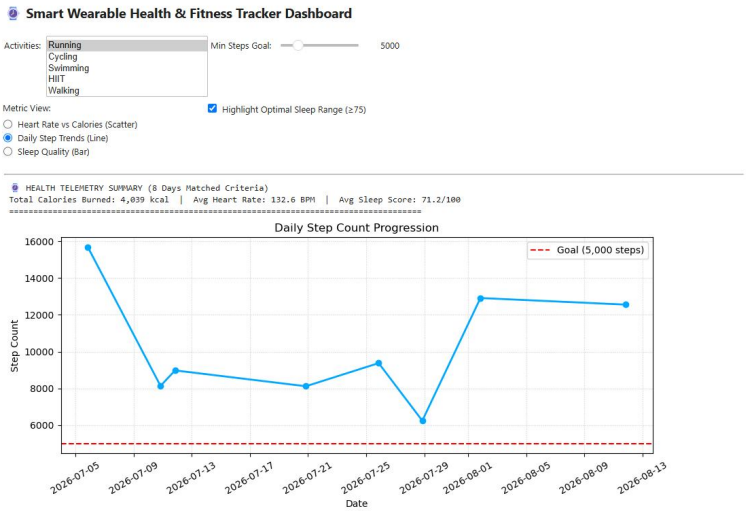
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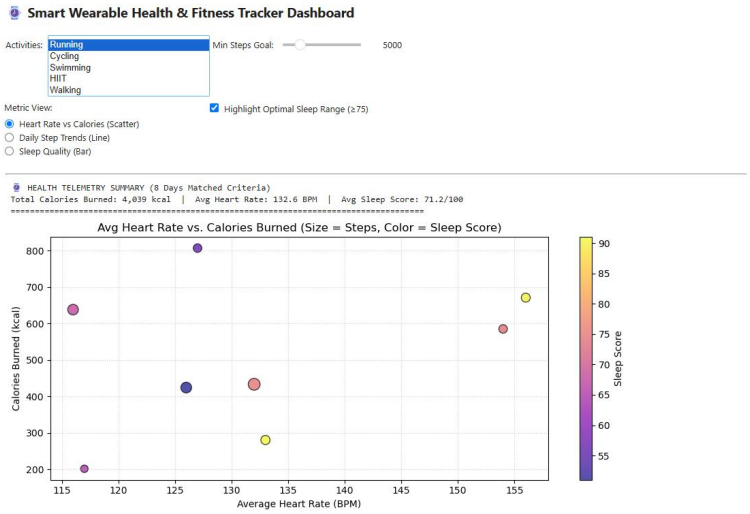
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CONCLUSION

Thus the code for health and fitness dashboard was successfully executed and visualized clearly.